

# Setting up your computer

NSDF-Data student guide 1 — Windows edition

Version 2 · 5 October 2026

## Tip

**Who this is for.** You are starting from little or no computing experience. You will paste some commands into a terminal and read the answers. You do **not** need to know any programming. Work through the steps in order. Each step ends with a **Checkpoint**: if it does not match, stop there.

## Before you start

- **Time:** about one hour. Most of it is waiting for downloads and installers.
- **You need:** an internet connection, the password for your computer, and about 3 GB of free disk space.
- **Rule of thumb:** type or paste commands **exactly** as written. Capitals and spaces matter.
- **When something goes wrong:** do not guess. Copy the message, note the step you were on, and ask the project lead. See *If something goes wrong* at the end.

## Words you will meet

| Word                         | What it means                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Terminal</b>              | A window where you type commands and the computer answers in text.                                             |
| <b>Command</b>               | One line you type in the terminal, followed by <b>Enter</b> (or <b>Return</b> ).                               |
| <b>Folder</b> (or directory) | A place on your computer that holds files and other folders.                                                   |
| <b>Path</b>                  | The address of a folder, such as<br>C:\Users\YourName\Research.                                                |
| <b>Git</b>                   | A program that downloads and keeps track of versions of code.                                                  |
| <b>Repository</b> (repo)     | A project folder that Git tracks. Here, NSDF-Data.                                                             |
| <b>Python</b>                | The programming language the analysis is written in. You will not need to write Python to finish these guides. |

| Word                     | What it means                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------|
| <b>Conda / Miniforge</b> | A tool that installs Python and packages in separate, named <i>environments</i> .             |
| <b>Environment</b>       | A sealed set of programs and versions, so this project's software does not clash with others. |
| <b>Package</b>           | A piece of ready-made software (for example numpy).                                           |
| <b>Jupyter notebook</b>  | A document that mixes text, code and plots, run one piece ( <i>cell</i> ) at a time.          |
| <b>Branch</b>            | A separate line of work in Git, so your changes do not touch the main code.                   |

## Step 1. Open a terminal

Click the **Start** button, type Miniforge Prompt, and press **Enter**. A black window with white text opens. This is the terminal for everything in this guide.

In the Miniforge Prompt, **right-click** the window to paste. `Ctrl+V` often does not work there.

### Checkpoint

You see a line of text ending in a symbol such as `>`, `$` or `%`. Type `echo hello` and press Enter: the word `hello` should appear under it.

## Step 2. Install Git

Go to **git-scm.com**, choose the **Windows** download, and run the installer. Keep every default setting: click **Next** until you reach **Install**, then click **Install**.

### Checkpoint

In the terminal, type `git --version` and press Enter. You should see a line starting with `git version`.

## Step 3. Install Miniforge

Miniforge installs Python and the other packages the analysis needs. Its download file is **Miniforge3-Windows-x86\_64.exe**.

Go to **github.com/conda-forge/miniforge**, find the file **Miniforge3-Windows-x86\_64.exe**, and download it. Double-click it and follow the installer:

- choose **Just Me**;
- keep the default install folder;
- **leave the box “Add Miniforge to my PATH” unticked**. The Miniforge Prompt works without it.

When it finishes, close any open Miniforge Prompt and open a new one.

### Checkpoint

Open a **new** terminal and type `conda --version`, then press Enter. You should see conda followed by a version number.

### Careful

Windows may show a blue warning box when you run an installer (“Windows protected your PC”). Click **More info**, then **Run anyway**, only if you downloaded the file from the link in this guide.

## Step 4. Download the NSDF-Data code

Your project will live in a folder called Research. Create it, go into it, and download the code. Type each line and press Enter:

```
mkdir C:\Users\YourName\Research
cd C:\Users\YourName\Research
git clone https://github.com/villano-lab/NSDF-Data.git
cd C:\Users\YourName\Research\NSDF-Data
```

Downloading with `git clone` does not need a GitHub account. It makes a full copy of the project in a folder named NSDF-Data.

### Checkpoint

Type `dir` and press Enter. You should see NSDF-Data in the list.

## Step 5. Create your analysis environment

An environment is a sealed box of programs with fixed versions. We make one for this project, called `darkmatter_cli_env`. Type these lines one at a time, pressing Enter after each. The third line takes several minutes and prints a lot of text: wait for the prompt to come back.

```
conda create -n darkmatter_cli_env python=3.10 -y
conda activate darkmatter_cli_env
python -m pip install nsdf-dark-matter==0.3.0 nsdf-dark-matter-cli==0.3.1
python -m pip install numpy==2.2.6 matplotlib==3.10.7 h5py pytest ipykernel
```

### Careful

Do **not** use the file `environment.yml` from the project. It was made on a Mac for Mac only, and it will fail on other computers. The three lines above install the same versions.

### Tip

Every time you open a new terminal, run `conda activate darkmatter_cli_env` first. When it is active, the start of each line shows `(darkmatter_cli_env)`.

### Checkpoint

The last line of the output is a message, not an error, and the (darkmatter\_cli\_env) label is at the start of your prompt.

## Step 6. Check that everything works

First, the tests. They use made-up data, so they need no download. Type each line and press Enter:

```
conda activate darkmatter_cli_env
cd C:\Users\YourName\Research\NSDF-Data\python
python -m pytest -q
```

### Checkpoint

The last line says 52 passed (it can be a larger number later, which is also fine). If it says failed, stop and ask the project lead, and do not change any files.

Then, the data tool:

```
nsdf-cli version
```

### Checkpoint

It prints NSDF Dark Matter CLI: 0.5.0 (or a later version).

## Step 7. Let Jupyter notebooks use your environment

Jupyter is the notebook program you will use. Register your environment with it once:

```
python -m ipykernel install --user --name darkmatter_cli_env
```

### Checkpoint

It prints a line that starts with Installed kernelspec darkmatter\_cli\_env.

## Step 8. Get permission to save your work

You will not change the main project. You will save your work on your own **branch**, and for that the lead must give you access:

1. Make a free account at **github.com** if you do not have one.
2. Send your GitHub username to the project lead. They will add you to the project as a collaborator.
3. Once you have accepted the invitation, set your name and email for Git. Use your own details:

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

4. Create your personal branch. Use your own name, without spaces, in place of yourname:

```
cd C:\Users\YourName\Research\NSDF-Data
git checkout -b student-yourname
```

## Checkpoint

The terminal shows Switched to a new branch 'student-yourname'.

## Rules for keeping your work safe

- **Never work on master.** That is the shared main version. Work only on your student-branch.
- **Never commit data.** The raw files in the idx folder are large and not part of the project.
- **Do not run 07221203\_2025\_dump1\_noise.ipynb** unless the project lead says it is fine. It overwrites a shared archive file.
- **Save often**, and commit your work when a piece is finished (see the first analysis guide).

## You are done when

- ☐ Your terminal opens and `git --version` prints a version.
- ☐ `conda --version` prints a version, in a new terminal.
- ☐ The NSDF-Data folder is in `C:\Users\YourName\Research`.
- ☐ `python -m pytest -q` in the python folder ends with 52 passed.
- ☐ `nsdf-cli` version prints a version.
- ☐ Your Jupyter kernel `darkmatter_cli_env` is installed.
- ☐ You have a student-branch (after access is granted).

**Next:** *First analysis* (the guide for Windows).

## If something goes wrong

1. Read the **last line** of the message. It usually says what is wrong.
2. Close the terminal, open a new one, and run `conda activate darkmatter_cli_env` again. Many problems are just the wrong environment.
3. Copy the command you ran and the message it printed, and send them to the project lead. A screenshot is also fine.

## Where to get help

**Anthony Villano**, project lead: [anthony.villano@ucdenver.edu](mailto:anthony.villano@ucdenver.edu)

When you write, include the step number, the command you typed, and the last few lines the computer printed. A screenshot helps. Nobody will be annoyed by a question about a step that did not work.

## Session Info

Guide version 2, 5 October 2026, Windows edition. Written for beginners; please tell the project lead where a step was unclear, so the guide can be fixed.

## Key links

- Notes site: <https://villano-lab.github.io/NSDF-Data/>
- Repository: <https://github.com/villano-lab/NSDF-Data>
- Miniforge downloads: <https://github.com/conda-forge/miniforge>
- Git downloads: <https://git-scm.com>